Team: Disco Funk (70)
Members: Jonathan Previsch, Riley Monroe

http://classwork.engr.oregonstate.edu:7070

Disco Funk Record Database

Project Step 3 Final: Design UI Interface + DML SQL

Feedback

Feedback from draft 3:

Cole Seifert

- ***Does the UI utilize a SELECT for every table in the schema?*** **In other words, data from each table in the schema should be displayed on the UI. Note: it is generally not acceptable for just a single query to join all tables and displays them.**
    - Yes, it look like each table utilizes a SELECT query and displays it nicely in different pages.
- ***Does the UI implement an INSERT form for at least one table in the schema?*** **In other words, there should be UI input fields that correspond to at least one table.**
    - No, I do not see any form that would be linked to an INSERT query.
- ***Does the UI have at least one DELETE for any one entity?*** **In other words, is there a form/link/button that will allow the deletion of a row in at least one table?**
    - No, I do not see any form/link/button that would relate to a DELETE query in their website.
- ***Does the UI have at least one DELETE that will remove things from a M:M relationship?*** **In other words, if an order is deleted from the Orders table, it should also delete the corresponding rows from the OrderDetails table, BUT it should not delete any Products or Customers.**
    - No, I do not see any DELETE forms/buttons resolving in a DELETE query.
- ***Is there at least one UPDATE form in the UI for any one entity?*** **In other words, in the case of Products, can productName, listPrice, qtyOnHand, e.g. be updated for a single ProductID record?**
    - No, I do not see any UI component for where an UPDATE query would be.

- ***Is there at least one UPDATE form in the UI to modify an M:M relationship?* In other words, does the UPDATE allow the user to select a different foreign key value to update the intersection table with?**
    - No, I do not see any UPDATE component like a form or button in their UI.
- ***Do you have any other suggestions for the team to help with their HTML UI? For example, using AS aliases to replace obscure column names such as fname with First Name.***
    - I love the overall styling of your UI, especially the Home page. It's easy to locate where everything will be once implemented. I would suggest adding,, or fixing the Purchase ID column under Purchases. Either it was not implemented correctly in the test data, or the query is incorrect as I do not see that column filled with anything.
- **As a reviewer, clearly describe to what extent your feedback to the team was original (e.g. "all my work") or non-original (e.g. used AI tools per Code Citation Tips).**
    - All my own work

Jasmine Davis

Does the UI utilize a SELECT for every table in the schema? In other words, data from each table in the schema should be displayed on the UI. Note: it is generally not acceptable for just a single query to join all tables and displays them.

Almost - while all the pertinent tables in the schema have a corresponding table in the UI, the team smartly elected to simply have an “inventory” UI table instead of creating one for PurchasesInventory and SalesInventory separately.

Does the UI implement an INSERT form for at least one table in the schema? In other words, there should be UI input fields that correspond to at least one table.

No, there are not currently any UI input fields on any of the pages.

Does the UI have at least one DELETE for any one entity? In other words, is there a form/link/button that will allow the deletion of a row in at least one table?

No, data cannot currently be deleted through the UI on any of the tables.

Does the UI have at least one DELETE that will remove things from a M:M relationship? In other words, if an order is deleted from the Orders table, it should also delete the corresponding rows from the OrderDetails table, BUT it should not delete any Products or Customers.

No, there are no delete options for any entities, including M:M.

Is there at least one UPDATE form in the UI for any one entity? In other words, in the case of Products, can productName, listPrice, qtyOnHand, e.g. be updated for a single ProductID record?

No, there is not currently a way to update any of the entities in the db.

Is there at least one UPDATE form in the UI to modify an M:M relationship? In other words, does the UPDATE allow the user to select a different foreign key value to update the intersection table with?

No, there is not a way to update M:M relationships via the UI currently.

Do you have any other suggestions for the team to help with their HTML UI? For example, using AS aliases to replace obscure column names such as fname with First Name.

Your styling is excellent and a lot of fun! Everything appears to be aliased correctly. Since your tables are styled so nicely, you might consider on-page editing with an “edit” and “delete” option within your tables as opposed to implementing on separate pages.

As a reviewer, clearly describe to what extent your feedback to the team was original (e.g. "all my work") or non-original (e.g. used AI tools per Code Citation Tips).

I wrote this review myself!

Mael Pierre

Does the UI utilize a SELECT for every table in the schema? **In other words, data from each table in the schema should be displayed on the UI. Note: it is generally not acceptable for just a single query to join all tables and displays them.**

Yes each table uses a select.

Does the UI implement an INSERT form for at least one table in the schema? **In other words, there should be UI input fields that correspond to at least one table.**

No, there are no UI input fields on any pages.

Does the UI have at least one DELETE for any one entity? **In other words, is there a form/link/button that will allow the deletion of a row in at least one table?**

Data currently cannot be DELETE on any page.

Does the UI have at least one DELETE that will remove things from a M:M relationship? **In other words, if an order is deleted from the Orders table, it should also delete the corresponding rows from the OrderDetails table, BUT it should not delete any Products or Customers.**

Currently, no entity can be deleted, even things from an M:M relationship.

Is there at least one UPDATE form in the UI for any one entity? **In other words, in the case of Products, can productName, listPrice, qtyOnHand, e.g. be updated for a single ProductID record?**

No, currently you can update any entity.

Is there at least one UPDATE form in the UI to modify an M:M relationship? **In other words, does the UPDATE allow the user to select a different foreign key value to update the intersection table with?**

No, you currently can't update any entity from the M:M relationship via the UI.

Do you have any other suggestions for the team to help with their HTML UI? For example, using AS aliases to replace obscure column names such as fname with First Name.

I really like your UI, the color palett of your website is very nice and it feels like an autumn theme. Obviously some edit and delete buttons need to be added to make it complete but aside form that everything looks good.

This review is mine entirely.

Gene Azad

Hi Jonathan Previsch and Riley Monroe,

My reviews can be found below:

> ***Does the UI utilize a SELECT for every table in the schema?*** **In other words, data from each table in the schema should be displayed on the UI. Note: it is generally not acceptable for just a single query to join all tables and displays them.**
>
> - Generally yes. However, two intersection tables (SalesInventor / PurchasesInventory) were rolled into one rather than having their own displays.

Does the UI implement an INSERT form for at least one table in the schema? **In other words, there should be UI input fields that correspond to at least one table.**

- No, there is no UI form for an INSERT

Does the UI have at least one DELETE for any one entity? **In other words, is there a form/link/button that will allow the deletion of a row in at least one table?**

- No. I didn't see any delete action in any table.

Does the UI have at least one DELETE that will remove things from a M:M relationship? **In other words, if an order is deleted from the Orders table, it should also delete the corresponding rows from the OrderDetails table, BUT it should not delete any Products or Customers**

    - No. I would suggesting adding logic to SalesInventory / PurchasesInventory that deletes Sale/Puchases.

Is there at least one UPDATE form in the UI for any one entity? **In other words, in the case of Products, can productName, listPrice, qtyOnHand, e.g. be updated for a single ProductID record?**

- No I saw no update UI

Is there at least one UPDATE form in the UI to modify an M:M relationship? **In other words, does the UPDATE allow the user to select a different foreign key value to update the intersection table with?**

- No. I would suggest adding an edit line item on the Sales pages to let users change the inventoryID and quantity on SalesInventory.

Do you have any other suggestions for the team to help with their HTML UI? For example, using AS aliases to replace obscure column names such as fname with First Name.

- I would suggest adding Edit/Delete actions and Add button on the list pages.

As a reviewer, clearly describe to what extent your feedback to the team was original (e.g. "all my work") or non-original (e.g. used AI tools per Code Citation Tips).

- All my work

Actions based on Feedback: Draft 3:

Added functionality for INSERT, UPDATE, DELETE UI to all pages following project requirements and feedback from peers, including M:M deletion properties. This is the vast majority of comments and was somewhat expected. As such, by adding these functionalities, we have fixed everything according to feedback.

Feedback from draft 2:

Hana Steffen

Does the schema present a physical model that follows the database outline and the ER logical diagram exactly?

- Yes, there are eight entities total and the schema follows the same structure as the ER Diagram and clearly illustrates the database outline as a physical model.

Is there consistency in a) naming between overview, outline, ER and schema entity/attributes b) entities plural, attributes singular c) use of capitalization for naming?

- All entity names and attributes are in camelCase and follow the same naming convention used in the overview and outline. Entity names are in plural and attributes are in singular.

Is the schema easy to read (e.g. diagram is clear and readable with relationship lines not crossed)?

- The schema is overall easy to read and understand. Suppliers is crossed with Purchases and Customers is crossed with Sales, but I think this could be resolved by placing Suppliers to the left of Purchases and Customers to the right of Sales.

Are intersection tables properly formed (e.g. two FKs and facilitate a M:N relationship)?

- Yes, there are two intersection tables. SaleInventory has a PK and two FKs and facilitates the M:N relationship between Sales and Inventory. PurchasesInventory has a PK and two FKs and facilitates the M:N relationship between Purchases and Inventory.

Does the sample data suggest any non-normalized issues, e.g. partial dependencies or transitive dependencies?

- There are no partial or transitive dependencies and the data seems to follow 1NF, 2NF, and 3NF guidelines.

Is the SQL file syntactically correct? This can be easily verified by using PhPMyAdmin and your CS 340 database (do not forget to take backup of your own database before you do this!)

- The SQL file is correct. Upon sourcing the DDL.sql script, no errors were encountered.

In the SQL, are the data types appropriate considering the description of the attribute in the database outline?

- All attribute datatypes make sense and look appropriate to me.

In the SQL, are the primary and foreign keys correctly defined when compared to the Schema? Are appropriate CASCADE operations declared?

- Yes, all PKs and FKs look correctly defined and consistent with the schema. There are no CASCADE operations.

In the SQL, are relationship tables present when compared to the ERD/Schema?

- Yes, all FKs correctly reference their corresponding PKs. All relationships are defined and consistent with the ERD/schema.

In the SQL, is all example data shown in the PDF INSERTED?

- All example data is successfully INSERTED into the database and consistent with sample data shown in the PDF.

Is the SQL well-structured and commented (e.g. hand authored) or not (e.g. exported from MySQL)?

- The DDL.sql file is well-structured and hand authored with some comments describing SQL queries.

As a reviewer, clearly describe to what extent your feedback to the team was original (e.g. "all my work") or non-original (e.g. used AI tools per Code Citation Tips).

- All of my work is original and no AI was used.

Shichun Xu

- Does the schema present a physical model that follows the database outline and the ER logical diagram exactly?

Yes, mostly. But I noticed some minor discrepancies: the data type of Inventory.type is ENUM in the schema and database outline, whereas it is VARCHAR in the ERD; and the Songs.album and Songs.format don't exist in data outline, It would be better to maintain consistency.

- Is there consistency in a) naming between overview, outline, ER and schema entity/attributes b) entities plural, attributes singular c) use of capitalization for naming?

Yes, mostly. Entities are plural and follow PascalCase, while attributes are singular and use camelCase across the outline, ER, and schema. However, it appears that Inventory.type in the ER and schema is referred to as Inventory.format in the outline.

- Is the schema easy to read (e.g. diagram is clear and readable with relationship lines not crossed)?

Yes! The schema is very easy to read.

- Are intersection tables properly formed (e.g. two FKs and facilitate a M:N relationship)?

Yes. The two intersection tables, SalesInventory and PurchasesInventory are properly formed.

- Does the sample data suggest any non-normalized issues, e.g. partial dependencies or transitive dependencies?

There are no partial dependencies. There might be a minor transitive dependency in Songs, where artist and dateReleased could depend on Album, but keeping these attributes is reasonable to reduce complexity. Additionally, I am slightly confused about Songs.format and Inventory.format, as they seem to represent the same data—unless there is a specific consideration I am missing?

- Is the SQL file syntactically correct? This can be easily verified by using PhPMyAdmin and your CS 340 database (do not forget to take backup of your own database before you do this!)

Yes. I can easily import the sql file into my database without any syntax errors.

- In the SQL, are the data types appropriate considering the description of the attribute in the database outline?

Yes, mostly. The only difference is in data size, such as Customers.name being varchar(50) in the outline but varchar(100) in the schema and SQL. It's a minor issue, but maintaining consistency is preferable.

- In the SQL, are the primary and foreign keys correctly defined when compared to the Schema? Are appropriate CASCADE operations declared?

The primary and foreign keys are properly defined; however, CASCADE operations are missing.

- In the SQL, are relationship tables present when compared to the ERD/Schema?

Yes, they are perfectly aligned.

- In the SQL, is all example data shown in the PDF INSERTED?

Yes, all the data has been inserted. However, there appears to be a discrepancy in Inventory.InventoryID between the SQL and the PDF. It’s also better to maintain consistency.

- Is the SQL well-structured and commented (e.g. hand authored) or not (e.g. exported from MySQL)?

The SQL is manually written, featuring a clear structure and comments.

All works are original.

Clay Coffman

- **Does the schema present a physical model that follows the database outline and the ER logical diagram exactly?**
    - Yes

- **Is there consistency in a) naming between overview, outline, ER and schema entity/attributes b) entities plural, attributes singular c) use of capitalization for naming?**
    - good job with consistency in naming conventions. All looks good.
    - Only think that caught my eye was using invoiceDate/invoiceAmount in the Sales table. Why not just make it salesDate and salesAmount? I understand the distinction between "sales" as a concept and an invoice, but maybe it makes more sense to name the table "Invoices" and then have views or reports downstream that a sales org would consume. Probably fine as is just something I thought worth calling out to think about. Also, it looks like in the pdf the id is invoiceId but in your ERD it's salesID.
- **Is the schema easy to read (e.g. diagram is clear and readable with relationship lines not crossed)?**
    - yes
- **Are intersection tables properly formed (e.g. two FKs and facilitate a M:N relationship)?**
    - yes
- **Does the sample data suggest any non-normalized issues, e.g. partial dependencies or transitive dependencies?**
    - it seems like format on the Songs table and type on the Inventory table might be redundant?
- **Is the SQL file syntactically correct? This can be easily verified by using PhPMyAdmin and your CS 340 database (do not forget to take backup of your own database before you do this!)**
    - yes
- **In the SQL, are the data types appropriate considering the description of the attribute in the database outline?**
    - yes
- **In the SQL, are the primary and foreign keys correctly defined when compared to the Schema? Are appropriate CASCADE operations declared?**
    - there are no CASCADE operations in the DDL. probably worth revisiting this and adding some.

- **In the SQL, are relationship tables present when compared to the ERD/Schema?**
    - yes
- **In the SQL, is all example data shown in the PDF INSERTED?**
    - yes
- **Is the SQL well-structured and commented (e.g. hand authored) or not (e.g. exported from MySQL)?**
    - yes

All my own work

Actions based on Feedback: Draft 2:

Overall feedback was positive, with little to no change needed. There were a few minor issues brought up by the feedback, but in discussing it ourselves, we decided to stick with the version that we had; not only was it less work for us, but the changes suggested were a matter of opinion. We reserve the right to change at a later date, however.

Feed back from draft 1:

Michael Fitzgibbon

Does the overview describe what problem is to be solved by a website with DB back end? If yes, summarize. If not, what changes would better support describing the problem to be solved?

Yes, the overview describes a music store in need of a back end driven website that will help facilitate expansion of their operations.

Does the overview list specific facts? If yes, summarize what the facts illustrate about the proposed DB solution. If not, what facts would better support illustrating the scope and scale of the proposed DB solution?

Yes, the overview does list expected customers and the expected restock requirements, but it could be improved by showcasing currently managed inventory and average inventory sales per customer.

Are at least four entities described, and does each one represent a single idea to be stored as a list? If yes, summarize. If not, based on the course material, what changes can you suggest to improve?

Yes, the entities listed are the Customers, Purchases, Supplier, and Inventory

Does the outline of entity details describe the purpose of each, list attribute datatypes and constraints, and describe relationships between entities? If yes, summarize. If not, based on the course material, what changes can you suggest to improve?

The function of each entity, as well as the purpose of its fields, are clearly explained in the outline. They also have logical restraints. One improvement I would make is to make less use of abbreviations for various terms, such as NOT NULL or AUTO INCREMENT for readability of the outline.

Are 1:M relationships correctly formulated? Is there at least one M:M relationship? Does the ERD present a logical view of the database? If yes, summarize. If not, based on the course material, what changes can you suggest to improve?

All relationships appear to be correctly formulated and the project fulfills the M:N relationship requirement. ERD makes sense.

Is there consistency in a) naming between overview and entity/attributes b) entities plural, attributes singular c) use of capitalization for naming? If yes, summarize. If not, based on the course material, what changes can you suggest to improve?

This could be improved. Table names change from plural to singular, Sales.InvoiceID is capitalized incorrectly, Customers.phone Number should be Customers.phoneNumber to keep with standards of the rest of the database. Otherwise, good design overall.

All views expressed in this review were created and shared by me alone, no outside assistance was provided.

Tony Miglets

1) **Does the overview describe what problem is to be solved by a website with DB back end? If yes, summarize. If not, what changes would better support describing the problem to be solved?**

The summary does a very good job of describing the problem that it solves. The group wants to create a database for a vintage record company that links customers, purchases, suppliers, inventory, and sales.

2) **Does the overview list specific facts? If yes, summarize what the facts illustrate about the proposed DB solution. If not, what facts would better support illustrating the scope and scale of the proposed DB solution?**

The summary also provides plenty of facts to support their reasoning for creating a database. They have roughly 1000 customers and 300 orders from suppliers. They're also a national company, planning to ship across the country. Some additional information, such as sales made per year, could also emphasize the need for a database.

3) **Are at least four entities described, and does each one represent a single idea to be stored as a list? If yes, summarize. If not, based on the course material, what changes can you suggest to improve?**

There are at least four entities, and each one represents one idea. They have customers, sales, purchases, supplier, inventory, salesinventory, and purchasesinventory.

4) **Does the outline of entity details describe the purpose of each, list attribute datatypes and constraints, and describe relationships between entities? If yes, summarize. If not, based on the course material, what changes can you suggest to improve?**

The outline provides a description of each entity as well as attribute datatypes. Relationships are also described. Customers has a 1:M relationship with sales implemented with customerID as a foreign key inside of sales. Sales has a M:M relationship with Inventory. Purchases has a M:M relationship with inventory. Supplier has a 1:M relationship with purchases.

5) **Are 1:M relationships correctly formulated? Is there at least one M:M relationship? Does the ERD present a logical view of the database? If yes, summarize. If not, based on the course material, what changes can you suggest to improve?**

All 1:M relationships are correctly formulated. There are two M:M relationships that are correct and logical. They also included information on the intersection tables used to describe these relationships.

6) **Is there consistency in a) naming between overview and entity/attributes b) entities plural, attributes singular c) use of capitalization for naming? If yes, summarize. If not, based on the course material, what changes can you suggest to improve?**

They have fairly consistent naming conventions in entities and attributes. I found one error in the Sales InvoiceID. This is the only one that didn't keep a consistent convention from the rest.

7) Originality

This is all my own work.

Charlie Tolls

Hi! I like your project idea and think your draft has a nice layout so far.

- **Does the overview describe what problem is to be solved by a website with DB back end?**
    - Yes, the overview describes the problem which is to create a website to expand and keep track of operations. It describes how the database will be used to track purchases from suppliers and sales to customers with various entities and transaction tables.
- **Does the overview list specific facts?**
    - The overview lists specific facts such as 1000 orders from customers and 300 orders to suppliers, however there could be room for additional facts that relate to the project goal and database scope. For example, "10,000+ different inventory items" (records, cassettes, etc) or "plans to make $50,000 in sales in one year".
- **Are at least four entities described, and does each one represent a single idea to be stored as a list?**
    - 3 tangible entities are mentioned (customers, suppliers, inventory) as well as sales and purchases (I'm unsure if these two are considered entities or transactional entities in this course). All entities represent a single idea to be stored as a list.
- **Does the outline of entity details describe the purpose of each, list attribute datatypes and constraints, and describe relationships between entities?**
    - Yes, all of these details are thoroughly listed in the outline of entity details. The relationships appear to be listed correctly and even the intersection tables are elaborated on.
- **Are 1:M relationships correctly formulated? Is there at least one M:M relationship? Does the ERD present a logical view of the database?**
    - Yes, the 1:M and M:M relationships appear correctly formulated and the ERD presents a logical and neatly formatted view with inventory being the central table, then Customers and Sales on one side and Suppliers and Purchases on the other.
- **Is there consistency in a) naming between overview and entity/attributes b) entities plural, attributes singular c) use of capitalization for naming? If yes, summarize. If not, based on the course material, what changes can you suggest to improve?**
    - This is one area that needs a bit of tweaking in the final draft. I noticed a couple instances where the first character of an attribute was capitalized in one table and uncapitalized in another. Also Supplier entity should be plural.

All my work

Gannon Strand

I believe your overview does indeed describe the problem that your database will solve, managing invoices that customers make, as well as ordering supplies from suppliers.

Your overview also contains specific facts, such as your expected number of orders being 1000, and your 300 orders to restock from your suppliers. An additional tidbit I would like to see is how many orders you expect to receive per year, per month, or if it indeed is only ever going to serve 1000 orders and then stop.

Yes, at least 4 entities are described in your database. And I think each one represents a single idea to be stored in a list. You have got your customers as an entity, sales as an entity, supplier as an entity, inventory as an entity, along with some intersection tables.

Each of your entities does have a purpose listed, attributes, and constraints. Your customer table has a primary key, customerID, autoincrementing, not null, and unique. One thing for consistency, you don't need your primary key if it's an integer and auto increments, you don't need to specify unique, which you don't in some places, but in other places you specify unique. You also have other attributes, such as customer name as a varchar(50), as well as several others listed. There are also relationships described for each, such as a M:M relationship between inventory and sales.

I believe your one-to-many relationships are correctly formatted. And you do have at least one M:M relationship! Your ERD presents a mostly logical view of your database. My only comment is that the line between SalesInventory and Inventory is not super clear.

There are a couple of things with consistency between naming that I wanted to point out. First off, your Supplier entity should probably be Suppliers, as you would have multiple, and that would more closely align with your other entities, Customers, Sales, etc. In your description, you have Phone Number as two words, when they should be combined as one to match everything else (you did correctly do this in your ERD diagram, just not in your description). Everything else seems pretty solid.

All my own work.

Neil King

+ Great use of numeric facts in your description

+ I would consider figuring out where you could reduce the number of M:N relationships while not required it may be more manageable

Actions based on Feedback: Draft 1:

Adding facts that would better support illustrating the scope and scale of the proposed DB solution. Put in info on current sales and inventory while also giving more reason for the database-backed website. Also included interest in eventual expansion through the database.

- showcasing currently managed inventory and average inventory sales per customer
- Some additional information, such as sales made per year, could also emphasize the need for a database.
- there could be room for additional facts that relate to the project goal and database scope. For example, "10,000+ different inventory items" (records, cassettes, etc) or "plans to make $50,000 in sales in one year".
- An additional tidbit I would like to see is how many orders you expect to receive per year, per month, or if it indeed is only ever going to serve 1000 orders and then stop.

Improve on details in outline of entity. Unabbreviated the terms such as NN to NOT NULL to increase readability. Removed some of the UNIQUE/NOT NULL attributes that weren't strictly necessary such as on the PRIMARY KEYs which automatically have those attributes. Also added some more details to other attributes.

- One improvement I would make is to make less use of abbreviations for various terms, such as NOT NULL or AUTO INCREMENT for readability of the outline.
- One thing for consistency, you don't need your primary key if it's an integer and auto increments, you don't need to specify unique, which you don't in some places, but in other places you specify unique

Fixing up ERD/Relationships and naming conventions. Fixed the error with invoiceID being capitalized as InvoiceID. Changed Supplier to Suppliers to match the rest of the table names. Choose to make them all plural instead of singular based on the recommendations in the instructions. Made the connections in the ERD clearer. Got rid of spaces in names such as phone Number to phoneNumber. We decided not to reduce the number of M:N relationships at this time, but we will keep it in mind if things become too complicated and we need to restructure.

- Table names change from plural to singular, Sales.InvoiceID is capitalized incorrectly, Customers.phone Number should be Customers.phoneNumber to keep with standards of the rest of the database.
- I found one error in the Sales InvoiceID. This is the only one that didn't keep a consistent convention from the rest.
- I noticed a couple instances where the first character of an attribute was capitalized in one table and uncapitalized in another. Also Supplier entity should be plural.
- My only comment is that the line between SalesInventory and Inventory is not super clear.
- First off, your Supplier entity should probably be Suppliers, as you would have multiple, and that would more closely align with your other entities, Customers, Sales, etc. In your description, you have Phone Number as two words, when they should be combined as one to match everything else (you did correctly do this in your ERD diagram, just not in your description).
- + I would consider figuring out where you could reduce the number of M:N relationships while not required it may be more manageable

a) Disco Funk Records Database Design Document

Disco Funk is a music store focused on the acquisition and distribution of vintage physical media, such as cassettes and Records. Using a paper filing system, it is becoming more difficult to properly track sales, purchases, and the increasing inventory of over 1000 current items has made it difficult to increase sales per year by more than 400. To expand operations, they've decided to set up a website to help them distribute music across the country, so that more people can enjoy authentic vintage music. The website is expected to receive orders from roughly 1000 customers over the next year, and expect to restock by making 300 orders from suppliers. Using a database will also allow for further continued growth year after year. The database attached to this website will record Invoices made by Customers based on the InventoryID of the item as well as Purchases made from Suppliers to increase items in the Inventory. This document outlines the details of the tables and relations needed to get the website online as well as sets the stage for future expansion.

b) Description and details of entities in database

Customers: A record of which customers have purchased from the store

- **customerID:** INT, AUTO INCREMENT, PRIMARY KEY
- **name**: VARCHAR (50), NOT NULL
- **address:** VARCHAR (100)
- **phoneNumber:** VARCHAR(15), UNIQUE
- **email:** VARCHAR (50), UNIQUE

Relationship: a 1:M relationship between Customers and Sales is implemented with customerID as a Foreign Key inside of Sales

Sales: A record of sales transactions made by customers.

- **salesID:** INT, AUTO INCREMENT, PRIMARY KEY
- **customerID:** INT, NOT NULL, FOREIGN KEY
- **invoiceDate:** DATETIME, NOT NULL, DEFAULT CURRENT_TIMESTAMP
- **invoiceAmount:** DECIMAL, NOT NULL
- **paymentType:** VARCHAR (20)
- **status:** ENUM('pending', 'completed', 'cancelled')

Relationship: a M:M relationship between Sales and Inventory through an intersection table. This is implemented with SalesID as a Foreign Key inside of SalesInventory.

Purchases: A record of purchases from suppliers.

- **purchaseOrderID:** INT, AUTO INCREMENT, PRIMARY KEY
- **supplierID:** INT, NOT NULL, FOREIGN KEY
- **purchaseDate:** DATETIME, NOT NULL, DEFAULT CURRENT_TIMESTAMP

- **totalCost:** DECIMAL
- **status:** ENUM('pending', 'completed', 'cancelled')

Relationship: a M:M relationship between Purchases and Inventory through an intersection table. This is implemented with PurchasesID as a Foreign Key inside of PurchasesInventory.

Suppliers: A record of suppliers who restock the inventory

- **supplierID:** INT, AUTO INCREMENT, PRIMARY KEY
- **name:** VARCHAR(50), NOT NULL
- **address:** VARCHAR (100)
- **email:** VARCHAR (50), UNIQUE
- **phoneNumber:** VARCHAR (15), UNIQUE

Relationship: a 1:M relationship between Supplier and Purchases is implemented with SupplierID as a Foreign Key inside of Purchases.

Inventory: A record of the inventory

- **inventoryID**: INT, AUTO INCREMENT, PRIMARY KEY
- **songID:** int, NOT NULL, FOREIGN KEY
- **type:** ENUM, ('Vinyl', 'CD', 'Cassette'), NOT NULL
- **quantity:** INT, NOT NULL, DEFAULT 0
- **price:** DECIMAL (10, 2)

Relationships:

- a M:M relationship between Inventory and Sales through an intersection table. This is implemented with InventoryID as a Foreign Key inside of SalesInventory.
- a M:M relationship between Inventory and purchases through an intersection table. This is implemented with InventoryID as a Foreign Key inside of PurchasesInventory.
- A M:1 relationship between Inventory and Songs is implemented with InventoryID as a Foreign Key inside of Songs

Songs: A record of song information

- **songID**
- **title:** VARCHAR (100)
- **genre:** VARCHAR(50), NOT NULL, FOREIGN KEY
- **artist:** VARCHAR (100), NOT NULL
- **album:** VARCHAR(50), NOT NULL
- **dateReleased:** YEAR
- **Format:** INT, NOT NULL

Relationships:

- M:1 relationship between Songs and Inventory is implemented with songID as a Foreign Key inside of Inventory

SalesInventory: An intersection table between Sales and Inventory.

- **salesInventoryID:** INT, AUTO INCREMENT, PRIMARY KEY
- **salesID**: INT, NOT NULL, FOREIGN KEY
- **inventoryID:** INT, NOT NULL, FOREIGN KEY
- **quantity:** INT, NOT NULL, DEFAULT 0

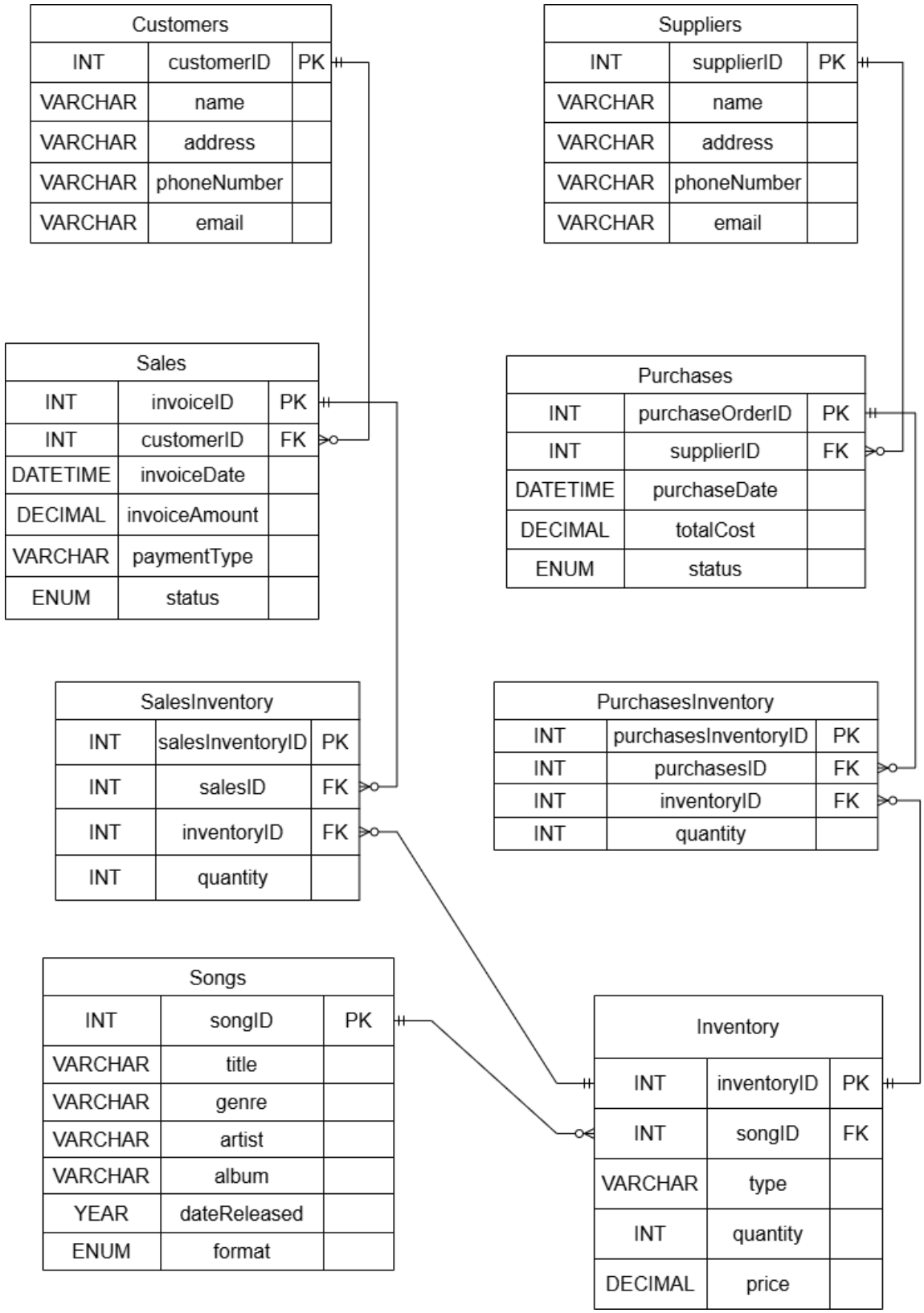
  Relationships: as an intersection table, forms part of a m:m with Sales and Inventory

PurchasesInventory: an intersection table between Purchases and Inventory.

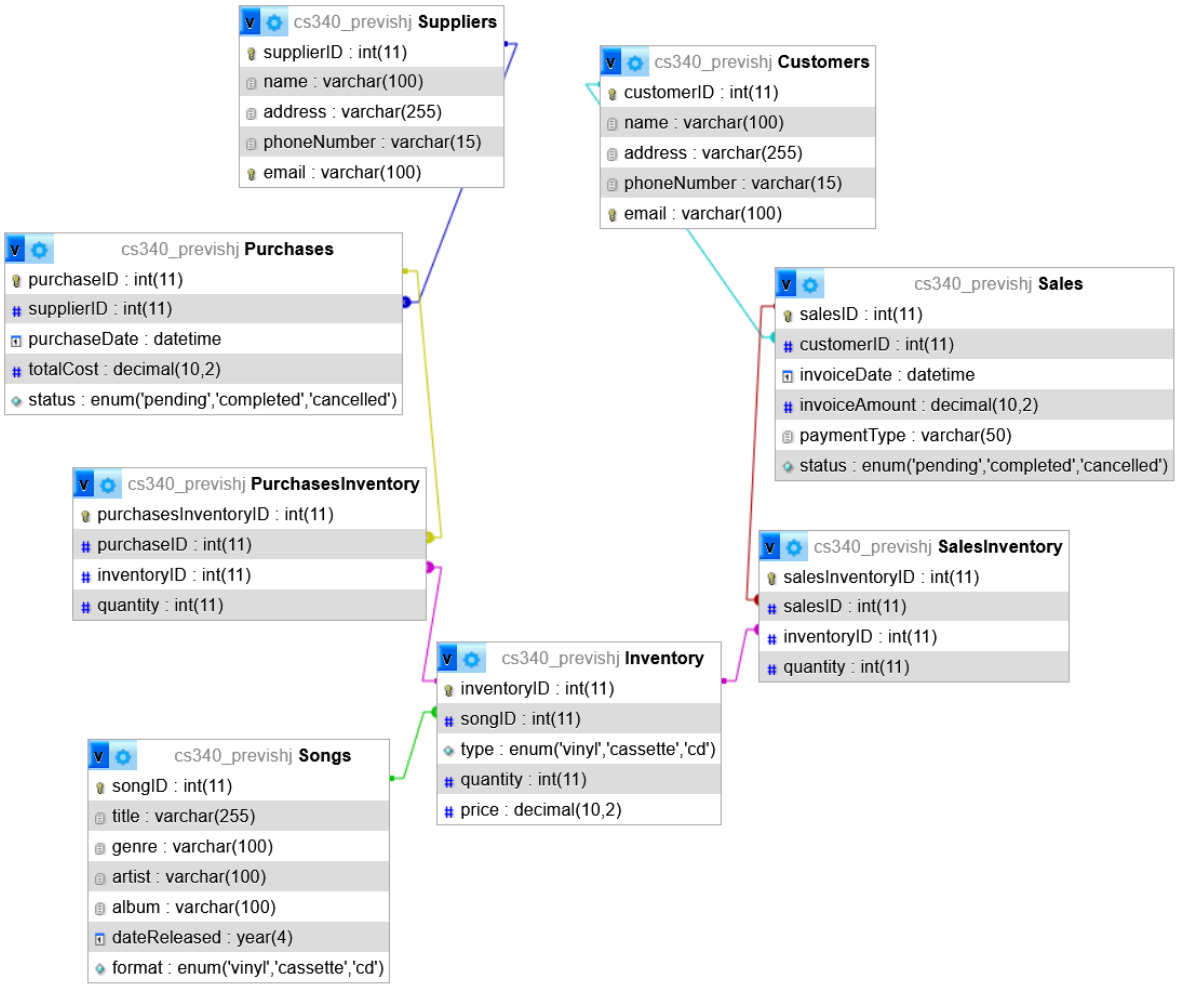
- **purchasesInventoryID:** INT, AUTO INCREMENT, PRIMARY KEY
- **purchasesID**: INT, NOT NULL, FOREIGN KEY
- **inventoryID:** INT, NOT NULL, FOREIGN KEY
- **quantity:** INT, NOT NULL, DEFAULT 0

  Relationships: as an intersection table, forms part of a m:m with Purchases and Inventory

Entity-Relationship Diagram: Draft



Database Schema:



Example Data:

Please note that all data is made up, and any similarities to real people or places is coincidental.

Customers:

Customers				
customerID	**name**	**address**	**phoneNumber**	**email**
1268	John Mamn	742 Evergreen Terrace, Springfield, IL 62701	(217) 555-1268	john.mamn@email.com
1269	Tom Redding	589 Pine Street, Boston, MA 02108	(617) 555-1269	tom.redding@mail.com
1270	Mary Marshal	1024 Cherry Lane, Denver, CO 80202	(720) 555-1270	mary.marshal@inbox.com
1271	Isaac Brown	1268 DoaksFerry Road NW, Salem, OR 97304	(304) 231-4655	isaacsmb@icloud.com

Sales:

Sales					
salesID	customerID	invoiceDate	invoiceAmount	paymentType	Status
4213	1268	10/12/25	$33.14	Visa	completed
4214	1269	10/12/25	$52.42	MasterCard	pending
4215	1270	10/13/25	$42.42	Cash	pending
4216	1271	10/14/25	$39.84	RoBucks	cancelled

SalesInventory (Intersection Table)**:**

SalesInventory			
SalesInventoryID	salesID	inventoryID	quantity
4213	4213	2	2
4214	4214	124	10
4215	4215	1	5
4216	4216	87	3

Suppliers:

		Suppliers		
supplierID	name	address	phoneNumber	email
624	Josh Bell	1847 Maple Street, Austin, TX 78701	(512) 555-0124	josh.bell@supplier.com
625	Dragon Joshington	2901 Oak Avenue, Portland, OR 97204	(503) 555-0625	dragon.j@dragonindustries.com
626	Mike Wazowski	456 Scare Floor Drive, Monstropolis, CA 90210	(626) 555-062	m.wazowski@monstersinc.com
627	Tony Romo	105 Kimbal Crossing, Kimball, TN 37347	(214) 555-062	tony.romo@romofootball.com

Purchases:

		Purchases		
purchaseID	supplierID	purchaseDate	totalCost	status
2259	624	10/1/25	$456.34	pending
2260	625	10/1/25	$229.77	cancelled
2261	626	10/1/25	$161.33	completed
2262	627	10/1/25	$983.31	completed

PurchasesInventory (intersection table):

		PurchasesInventory	
purchasesInventoryID	purchasesID	inventoryID	quantity
2259	2259	1	25
2260	2260	2	23
2261	2261	87	17
2262	2262	123	82

Songs:

Songs						
SongID	title	genre	artist	Album	dateReleased	format
1	"Stayin' Alive"	Disco	Bee Gees	Saturday Night Fever	1977	1
5	"Le Freak"	Disco	Chic	C'est Chic	1978	1
87	"Boogie Oogie Oogie"	Disco	A Taste of Honey	A Taste of Honey	1978	1
123	"I Feel Love"	Disco	Donna Summer	I remember Yesterday	1977	3

Inventory:

Inventory				
InventoryID	SongID	type	quanity	price
1	1	cassette	23	$12.34
2	1	vinyl	45	$15.42
87	5	vinyl	34	9.99
123	87	vinyl	67	$13.01
124	123	vinyl	187	$16.94

Citations

https://www.w3schools.com/sql/sql_datatypes.asp

Harrington, Jan L. *Relational Database Design and Implementation*. Morgan Kaufmann/Elsevier, 2016.